

Efficient Catalytic Upgrading of Ethanol to Higher Alcohols via Inhibiting C–C Cleavage and Promoting C–C Coupling over Biomass-Derived NiZn@NC Catalysts

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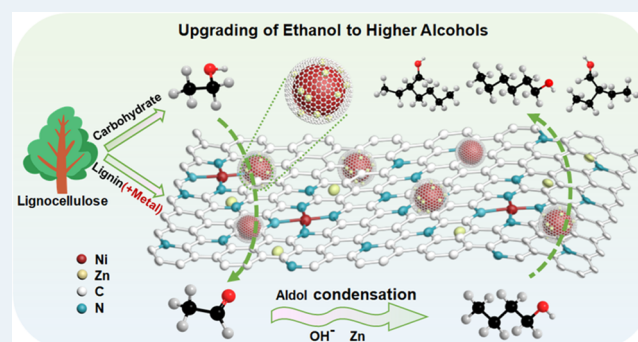
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ABSTRACT: The conversion of renewable bioethanol into high-energy-density higher alcohols has become essential for meeting the increasing global demand to achieve carbon neutrality. In this study, Zn- and nitrogen-codoped Ni-based lignin-derived carbon catalysts (NiZn@NC) were prepared by solvent volatile self-assembly and *in situ* reductive carbonization using pulp and paper waste stream alkali lignin as the carbon source. Lignin amphipathic derivatives with –COOH and –NH₂ groups would coordinate with metal ions to form a stable lignin–metal framework; thus, the lignin-derived carbon layer disperses the NiZn bimetallic catalyst and prevents from corroding. At an amination reagent/lignin mass ratio of 1:2, an ethanol conversion of 75.2% and a high alcohol yield of 41.7% were achieved over the Ni₂₀Zn₁@NC catalyst.

Experimental results and density functional theory calculations showed that Zn doping improved the electronic environment and defect structures of metallic Ni and carbon carrier, which effectively inhibited C–C cleavage and suppressed the byproduct formation, such as methane. Thereby, the synergetic effect between Ni and Zn facilitated the efficient conversion of aqueous ethanol into higher alcohols by the Guerbet reaction. This work provides a strategy of *in situ* pyrolytic doping and stabilizing of renewable biomass macromolecules as the frameworks for the construction of highly active and cost-efficient catalysts for ethanol upgrading.

KEYWORDS: ethanol coupling, higher alcohols, Guerbet reaction, NiZn bimetallic catalysts, lignin-derived carbon



INTRODUCTION

Because of the increasing consumption of fossil fuels, the search for sustainable, affordable, and green alternative fuels (or fuel sources) has become a widespread topic of interest.^{1,2} Bioethanol produced by the fermentation of lignocellulose biomass has shown promise as bio-gasoline, either in its pure or blended form.^{3,4} However, the disadvantages of bioethanol, such as its high solubility in water, low energy density, and tendency to corrode, considerably limit its applications as a fuel. Therefore, the conversion of bioethanol to higher alcohols, with properties similar to those of petrol, can be an effective solution.^{5,6} Higher alcohols are immiscible in water, more energy dense, and less corrosive to engines and other fuel infrastructure, thereby proving to be promising bio-derived fuels.^{7–9} Synthesizing higher alcohols via non-petrochemical techniques, such as acetone–butanol–ethanol fermentation, remains challenging as obtaining the desired C₄₊ alcohols is difficult.⁴ The conversion of fermented bioethanol to higher alcohols via the C–C coupling reaction shows great potential.

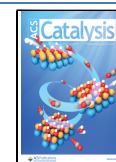
The traditional mechanism of ethanol coupling involves dehydrogenation, aldol condensation, and hydrogenation, in

accordance with the mechanism of the Guerbet reaction.^{10,11} Ni-based catalysts are excellent dehydrogenation and hydrogenation agents, and are one of the most effective catalysts for ethanol coupling.^{12,13} Pang et al.¹⁴ prepared highly dispersed Ni-MgAlO catalysts where the metallic Ni sites improved ethanol dehydrogenation, achieving 55 and 85% selectivity to *n*-butanol and C₄–C₈ alcohols, respectively, at an ethanol conversion of 18.7%. Nezam et al.¹⁵ used Ni/La₂O₃/γ-Al₂O₃ catalysts to convert ethanol to higher alcohols at 230 °C with 75–80% selectivity. However, a high dehydrogenation rate and the ability to break C–C/C–O bonds by metallic Ni^{16–18} boost the conversion, leading to the formation of byproducts such as methane (CH₄), carbon monoxide (CO), and carbon dioxide (CO₂). The byproduct formation lowers the yields of

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higher alcohols. Zn doping is effective in suppressing carbon deposition and improving the electronic environment of metallic Ni to an electron-rich state, thereby increasing the number of defect structures on Ni-based catalysts.¹⁹ Zn-doped Ni-based catalysts can effectively inhibit C–C bond cleavage and CH₄ formation during hydrolysis, thereby enhancing the catalyst activity to obtain the target product(s) selectivity.^{20–23} Pan et al.²¹ reported the transfer charge on Ni to Zn in Ni–Zn alloys and other Ni–Zn compounds and that Ni–Zn compounds can significantly inhibit the C–C bond hydrogenolysis and methanation, thereby promoting hydrodeoxygenation of methyl laurate and improving the catalytic performance. Kumar et al.²² modified the electronic composition of a Ni catalyst by adding Zn to enrich Ni with electrons, effectively maintaining Ni in the metallic state, to improve the stability of the catalyst during the triple reforming of CH₄. Therefore, the hypothesis of Zn-doped Ni-based catalysts is proposed in this study for the efficient and highly selective catalytic conversion of ethanol to higher alcohols via the Guerbet reaction.

Lignin, a naturally occurring aromatic polymer, contains ~60% organic carbon, which has an irregular network structure of monomeric benzene propane, and is rich in functional groups (such as phenolic –OH, –OH, etc.).^{24,25} These abundant functional groups facilitate the modification of lignin through effective coordination and dispersion of metals by forming a lignin–metal framework and further *in situ* reduction and carbonization to form the metallic catalyst encapsulated and embedded by the carbon carrier, which can exploit the synergy between the metal and protective carbon layer to enhance the catalytic activity and stability.²⁶ Additionally, nitrogen *in situ* doping through lignin amination leads to product pyridinic-N and graphitic-N, as well as abundant edge-defect structures, that can improve the interaction and dispersion between the active metals and the carbon carrier.^{27–31}

In this study, Zn/N-codoped Ni-based bimetallic catalysts (NiZn@NC) were prepared by *in situ* reductive carbonization of lignin–metal complex precursors formed through the self-assembly of lignin biomolecules with metal ions. Zn/N codoping improved the microscopic morphologies of the pore structure and defect structures of the catalysts, resulting in the efficient conversion of ethanol to higher alcohols. Additionally, the effect of Zn doping on C–C and C–O bond cleavage was investigated by density functional theory (DFT) calculations, and the mechanism for converting aqueous ethanol to C₄₊ alcohols over lignin-derived carbon material catalysts has been proposed.

MATERIALS AND METHODS

Materials. Alkali lignin (AL), with an average molecular weight of 6400 g/mol, was provided by Shandong Longli Biotechnology Co., Ltd. Ni(NO₃)₂·6H₂O (98%), (CH₃COO)₂Zn (99%), NaOH (97%), formaldehyde (37 wt %), ethanol (≥99.5%), and dimethyl sulfoxide (99%) were supplied from Aladdin. Chloroacetic acid (98%) and diethylenetriamine (99%) were acquired from MacLean. All of the chemicals were applied without additional purification.

Preparation of Modified Lignin. AL was modified via carboxymethylation and amination to obtain a water-soluble lignin precursor. A typical synthesis was adding 10 g of AL to 20 g of NaOH (20 wt %) solution while stirring until all of the AL was dissolved. Then, 10 ml of deionized water was used to dissolve 6 g of monochloroacetic acid, and the mixture was

added dropwise to the lignin solution. A homogeneous solution was achieved after 10 min of continuous stirring. The reaction temperature was raised to 70 °C, and it was carried out under stirring for 90 min to gain the carboxymethylated lignin, which was followed by the Mannich reaction. Diethylenetriamine (5 g) was added dropwise to the abovementioned lignin solution and stirred for 10 min and 20.3 g of 37 wt % formaldehyde was added and reacted at 70 °C for 4 h under stirring. The slurry was cooled to 30 °C, and 20 wt % lignin derivative was prepared for the next step of the catalyst synthesis. To prepare modified lignin with various degrees of amination, the mass ratios of diethylenetriamine to the original lignin were adjusted to 1:4, 1:2, 3:4, and 1:1, and the mass of added formalin varied in the same proportion as the amount of added diethylenetriamine. The modified lignin samples were named 1/4-AL, 1/2-AL, 3/4-AL, and 1/1-AL, and the lignin sample modified via carboxymethylation was named as control-AL.

Catalyst Preparation. Sheet-like catalysts of graded porous nitrogen-doped bimetallic carbon materials were prepared via solvent volatilization self-assembly. Ni(NO₃)₂·6H₂O (1.82 g) and (CH₃COO)₂Zn (0.0574 g) were taken in two beakers separately and dissolved in deionized water (10 mL). Ni(NO₃)₂·6H₂O solution and (CH₃COO)₂Zn solution were slowly added to the 1/2-AL (18.20 g, 20 wt %) lignin precursor under continuous stirring until the formation of a dark brown precipitate. The molar ratio of Ni/Zn was 20:1, and that of metal/lignin was 12.9:1. The dark brown precipitate was stirred at room temperature for 2 h and was then additionally stirred at 100 °C until complete evaporation of water. The lignin–metal complex precipitate was dried at 90 °C and given the designation 1/2-AL-NiZn. The mixture was crushed into powder and pyrolyzed in a tube furnace in the N₂ environment. The temperature was raised at a rate of 3 °C/min to 150 °C, 2 °C/min to 350 °C, and lastly at a rate of 1.8 °C/min to 500 °C. The final temperature of 500 °C was maintained for 2 h. The prepared catalysts (NiZn@NC) were used to characterize and assess the catalytic activity. To investigate the reusability of the catalysts, they were centrifuged apart, washed three times with water and ethanol, and then dried overnight at 60 °C before reuse.

Catalyst Characterization. Fourier-transform infrared (FTIR) spectroscopy was carried out on an iS50R full-band research-grade infrared spectroscopy workstation (Thermo Fisher) with a test wavelength range of 4000–400 cm^{–1}. Powder X-ray diffraction (XRD) patterns were recorded in an X-ray diffractometer (TD-3500, Liaodong Radioactive Instrument, China) using Cu K α radiation. Nitrogen adsorption and desorption isotherms were investigated using a Micromeritics ASAP 2020 instrument. The catalysts were initially degassed at 200 °C for 12 h before the test, and then the isotherms were collected at 77 K. Electron paramagnetic resonance (EPR) spectroscopy (Bruker A300) was used to measure the unpaired electrons of the catalysts. To assess the surface chemical properties of the as-synthesized catalysts, X-ray photoelectron spectroscopy (XPS) was undertaken using a Thermo K-Alpha+ apparatus. The nickel and zinc contents of the catalysts were ascertained using inductively coupled plasma-atomic emission spectrometry (ICP-OES) in a Varian (720) apparatus. The surface morphology and microstructure were examined using a high-resolution field-emission scanning electron microscope (SEM) in a Zeiss Sigma 500 apparatus and transmission electron microscopy (TEM) in an FEI Tecnai G2 f20 s-twin

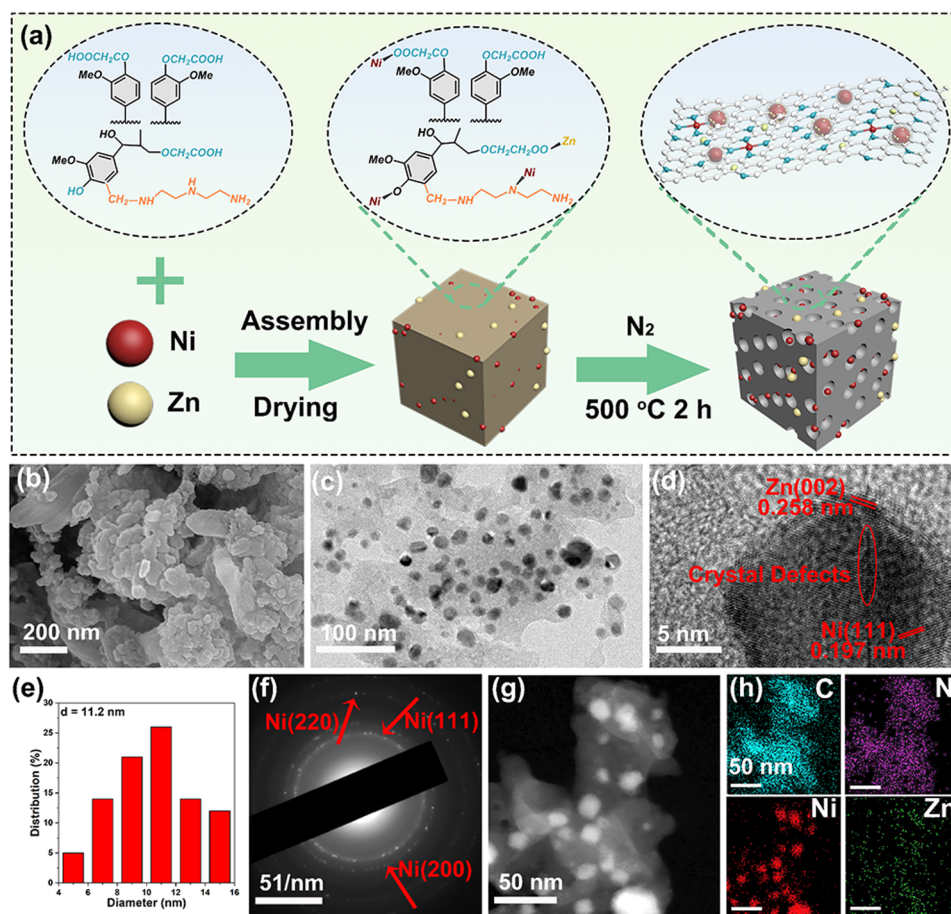


Figure 1. (a) Schematic of preparation, (b) SEM image, (c) TEM image, (d) HRTEM image, (e) particle size distribution, (f) SAED image, (g) HAADF-STEM image, and (h) EDS elemental map of C, N, Ni, and Zn for the Ni₂₀Zn₁@NC catalyst.

apparatus. The catalysts were first dispersed in ethanol using ultrasound before being placed dropwise onto a carbon film anchored by a copper grid. X-ray absorption fine structure (XAFS) spectroscopy was carried out using the Rapid XAFS1M (Anhui Absorption Spectroscopy Analysis Instrument Co., Ltd.) by transmission mode at 20 kV and 40 mA, and the Si (551) spherically bent crystal analyzer with a radius of curvature of 500 mm was used.

DFT Calculations. All DFT computations were conducted with the Perdew–Burke–Ernzerhof functional using the Vienna Ab initio Simulation Package.^{32,33} Due to a lower surface energy, the Ni catalyst was modeled as a (111) facet with a lattice constant of 3.51 Å. The decomposition of ethanol to methane was studied over Ni(111) using a 4×4 unit cell (Figure S1a), sampled with a $4 \times 4 \times 1$ γ -centered Monkhorst–Pack k -point mesh, whereas a 6×6 unit cell (Figure S1b) sampled with a $2 \times 2 \times 1$ k -point mesh was adopted for ethanol coupling studies over Ni(111).^{23,34} Similarly, the Zn(0001) surface was modeled using two cell sizes: 3×3 and 5×5 cells (Figure S1c,d), sampled with $4 \times 4 \times 1$ and $2 \times 2 \times 1$ k -point grids, respectively. To mimic the Ni₂₀Zn₁@NC catalyst used in the experiments, we also constructed a ZnNi(111) model by optimizing a Zn₈ cluster on the Ni(111) surface, and the most stable structure is shown in Figure S1e. For the ZnNi(111) model, a 6×6 unit cell sampled with a $2 \times 2 \times 1$ k -point mesh was adopted for all calculations. In all cases, the metal slabs consisted of four layers, with the bottom two layers being fixed in their bulk-

truncated positions. A vacuum space of 15 Å was inserted between successive slabs. The Kohn–Sham one-electron states were enlarged based on plane waves with a 400 eV energy cutoff. The dispersion interactions were described using the DFT-D3 method.³⁵ The forces on all atoms were thought to be convergent when they were less than 0.05 eV/Å. The transition states were located using the dimer approach and the climbing image nudged elastic band method.^{36–38}

Catalytic Evaluation. The ethanol coupling process of ethanol to higher alcohols was conducted in a reaction kettle (MS-50-316L, Anhui Kemi Machinery Technology Co., Ltd., Hefei, China) fitted with a stirring blade. The dosages of ethanol, water, catalyst, and NaOH in the autoclave were 15, 15, 0.5, and 1 g, respectively. To replace the air, hydrogen (H₂) was flushed through the autoclave four times. To bring the pressure in the autoclave down to atmospheric pressure, some of the H₂ was released. The reaction was conducted with 500 rpm stirring speed at 250 °C for 24 h. Regarding the completion, the reactor was cooled to ambient temperature, the reactor's gas was gathered in a bag during the depressurization procedure, and the liquid products were gathered and weighed for additional examination.

Product Analysis. The drainage technique was used to calculate the volume of gaseous products. The gaseous hydrocarbon products were detected using gas chromatography (GC9800, Kechuang, China) equipped with an FID detector and an HP-AL/S capillary column (50 m $\times$ 0.535 mm $\times$ 15 μ m). It was conducted using the following temperature

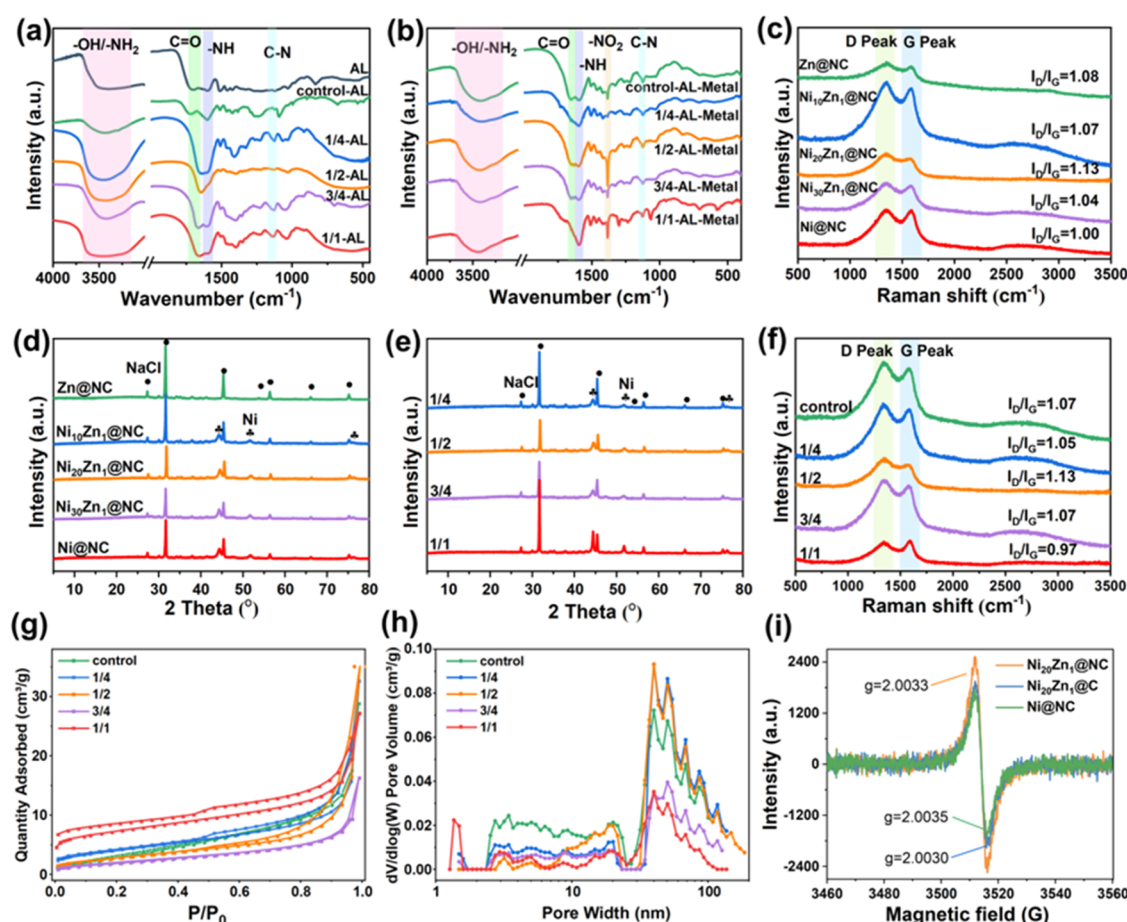


Figure 2. (a) FTIR spectra of alkali lignin, lignin modified with various degrees of amination and (b) lignin–metal complex precursors prepared from lignin modified with various amination ratios. (c) Raman spectra of NiZn@NC catalysts with various NiZn ratios, (d) XRD patterns of NiZn@NC catalysts with varied NiZn ratios, and (e) Ni₂₀Zn₁@NC catalyst prepared from varied amination ratios of lignin. (f) Raman spectra of Ni₂₀Zn₁@NC catalysts prepared from various amination ratios of lignin. (g) Nitrogen adsorption–desorption isotherms and (h) pore size distributions of Ni₂₀Zn₁@NC catalysts prepared from varied amination ratios of lignin. (i) EPR curves of different Ni-based catalysts.

program: an initial temperature of 60 °C for 5 min followed by a 20 °C/min ramp up to 200 °C. The gaseous CO, CO₂, and H₂ products were measured under the following conditions: beginning temperature of 60 °C for 5 min and ramping up to 120 °C at a rate of 20 °C/min using a TCD detector and a 5A Zeolite column (2 m × 4 mm × 2 mm). The liquid product underwent an aqueous and organic centrifugation process. For the purpose of determining the reaction's carbon balance, the samples were individually weighed. The aqueous and organic phase products were each filtered using a syringe filter before the liquid products were qualitatively analyzed by GC with mass spectrometry (GC-MS, Agilent, HP-INNOWax, Thermo Fisher Scientific TRACE 1300ISQ). The GC column was first heated at 50 °C for 3 min, then to 240 °C at a rate of 15 °C/min, and kept at 240 °C for 3 min. Using a GC system (Shimadzu 2010 Pro) equipped with a HP-INNOWax column (30 m × 0.32 mm × 0.50 μm), the liquid products were quantitatively evaluated using the internal standard technique, where the internal standard was 1-pentanol and the solvent was dimethyl sulfoxide.

The carbon balance is calculated using eq 1 depending on the amount of carbon present before and after the reaction. The carbon balances are often between 70 and 90% because of the partial loss of CO₂ brought by NaOH absorption, the volatile loss during the depressurization process, the loss

during the liquid gathering procedure, and the unidentified component. The conversion, yield, and selectivity were calculated depending on the quantity of moles of carbon (C %), according to eqs 2–4

$$\text{carbon balance (\%)} = \frac{\sum n - C_i}{n - C_{\text{EtOH}}} \quad (1)$$

$$\text{conversion (C\%)} = \frac{n_{\text{EtOH}} - n_{\text{unreacted EtOH}}}{n_{\text{EtOH}}} \quad (2)$$

$$\text{yield (C\%)} = \frac{n - C_i}{n_{\text{EtOH}}} \quad (3)$$

$$\text{selectivity (C\%)} = \frac{n - C_i}{\sum n - C_i} \quad (4)$$

RESULTS AND DISCUSSION

Catalyst Morphological and Structural Analysis. The Ni₂₀Zn₁@NC catalyst was prepared via solvent volatilization self-assembly and *in situ* reductive nitrogen-doped carbonization, as shown in Figure 1a. The alkali lignin was modified via carboxymethylation and amination to increase the N and O contents, that was verified through XPS spectra and elemental analysis as shown in Figure S2 and Table S1. The modified

lignin coordinated with metal ions to form lignin–metal complexes, which were further reduced and carbonized *in situ* to obtain the $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ catalyst. The microscopic morphology of the $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ catalyst is shown in Figure 1. The $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ catalyst has a lamellar thin layer and highly porous structure as shown in Figure 1b,c. The metallic component is encapsulated in the amorphous carbon material (Figure 1d), as proved by lattice fringes with a crystalline spacing of 0.197 nm, corresponding to the Ni(111) crystal plane in the XRD pattern. The lattice fringes with an interplanar spacing of 0.258 nm are attributed to the (002) crystal plane of Zn, indicating the doping of Zn into Ni nanoparticles, and the average particle size of Zn-doped Ni nanoparticles was 11.2 nm (Figure 1e).

The crystal defect structures in metal clusters, owing to the doping of Zn, can be observed in the HRTEM image. In addition, HAADF-STEM and EDS elemental mapping images (Figure 1g,h) show that Zn is uniformly distributed on the N-doped carbon material for its stronger coordination with amine groups, while Ni formed nanoparticles due to the strong Ni–Ni bonding. The percentage contents of Ni and Zn were 17.18 and 0.82%, respectively, determined by ICP-OES.

The FTIR profiles of raw alkali lignin, lignin modified by various degrees of amination, and lignin–metal complex precursors are shown in Figure 2a,b. The in-depth coordination between metals and lignin was explored by FTIR. The spectrum of alkali lignin (Figure 2a) shows a strong absorption band in the region of 3700–3200 cm^{-1} , indicating the presence of phenolic and alcoholic hydroxyl groups. The band at 1715–1660 cm^{-1} is attributed to the C=O group, and the absorption bands at 1610–1600, 1520–1500, and 1450–1420 cm^{-1} are ascribed to the skeletal vibrations of the benzene ring.^{39,40}

There are some significant variations in the FTIR profiles of modified lignin with different degrees of amination. The absorption band representing C=O at 1715–1660 cm^{-1} is significantly stronger because of the –COOH group introduced during carboxymethylation of lignin. The strong absorption bands at 1650–1590 cm^{-1} for 1/4-AL, 1/2-AL, 3/4-AL, and 1/1-AL are the shear vibrational bands of –NH₂ group. The vibrational band of –C–N at 1138 cm^{-1} indicates the successful grafting of amine group onto lignin.⁴¹

The FTIR spectra of the lignin–metal complex precursors prepared using different amine ratios are shown in Figure 2b. The absorption bands at 1380 cm^{-1} for the precursors with the lignin/amine reagent mass ratios of 1/2, 3/4, and 1/1 are significantly intense, indicating the presence of –NO₂ groups. This is because of the successful grafting of the modified lignin with the metallic Ni during the precursor synthesis and introduction of nickel nitrate, resulting in the appearance of a –NO₂ absorption band.⁴¹

XRD patterns of NiZn@NC prepared with varied Ni/Zn ratios and varied degrees of lignin amination modification are shown in Figure 2d,e. As shown in Figure 2d, all catalysts exhibit sharp reflections, indicating a more regular crystal structure of the prepared catalysts. The diffraction peaks at 44.4, 51.8, and 76.6° correspond to the (111), (200), and (220) crystallographic planes of metallic Ni, compatible with the standard card (JCPDS #04-0850)⁴² and the selected-area electron diffraction (SAED) of the $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ catalyst (Figure 1f), further demonstrating the presence of metallic Ni in the carbon-containing catalysts. XRD patterns of $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ prepared using modified lignin with various amination

ratios also exhibit reflections ascribed to metallic Ni as shown in Figure 2e, which also indicates the successful doping of metallic Ni. The diffraction peaks at 31.77, 34.42, 56.59, and 66.37° correspond to the crystal planes of ZnO that can be observed by local magnification in Figure S3.

The nitrogen adsorption–desorption isotherms and pore size distribution of $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ catalysts prepared using modified lignin with various degrees of amination are shown in Figure 2g,h. Hysteresis loops in the P/P_0 range of 0.4–1.0 are present in the nitrogen adsorption and desorption isotherms, corresponding to isothermal hysteresis loops of type H1.⁴³ This indicates the presence of a mesoporous network in the carbon produced from lignin, with a relatively broad range of pore size distribution and a more homogeneous particle size, which confirms the SEM and TEM results of the $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ catalyst, further demonstrating that metals are effectively dispersed by carboxymethylated and aminated lignin derivatives.²⁷ The pore size distribution diagram confirms that the pore channels of the catalysts have microporous–mesoporous–macroporous hierarchical multilevel pores, which would facilitate the adsorption of ethanol and intermediates on the catalyst and increase the mass-transfer rate.⁴⁴

To further investigate the graphitization state of the NiZn@NC catalyst and the extent of defects in the graphitic carbon layer, Raman spectroscopy was performed (Figure 2c,f). The characteristic bands near 1580 and 1350 cm^{-1} for the NiZn@NC catalysts are the G- and D-bands, respectively. It is generally accepted that the D-band is due to the defects in the carbon lattice and vibrations at the ends of disordered graphite planes, whereas the G-band is associated with vibrations of sp^2 -hybridized carbon in the graphite layers.^{45,46} The ratio of the Raman intensities of the D- and G-bands (I_D/I_G) has been widely used to analyze carbon species.

The I_D/I_G values of the catalysts with varied Ni/Zn ratios are shown in Figure 2c. The I_D/I_G values of Zn@NC , $\text{Ni}_{10}\text{Zn}_1\text{@NC}$, $\text{Ni}_{20}\text{Zn}_1\text{@NC}$, and $\text{Ni}_{30}\text{Zn}_1\text{@NC}$ catalysts are higher than those of the Ni@NC catalysts, establishing that Zn doping facilitates the formation of defects (also observed in HRTEM results). The catalyst exhibits the largest I_D/I_G value (1.13) when the Ni/Zn ratio is 20:1, indicating the presence of a large number of defective structures. The I_D/I_G values of $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ catalysts prepared with various lignin amination ratios are shown in Figure 2b. Compared with the control and 1/4 amination degree, the I_D/I_G ratios of $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ catalysts at 1/2 degrees of amination show the highest defect level ($I_D/I_G = 1.13$). Further increasing the amination degree would decrease the number of metal ions coordinated with lignin derivatives as shown in Figure 2b, which is consistent with the percentage of metals in the NiZn@NC catalysts shown in Table S2, thus decreasing the defects.

The effect of Zn and N doping on the defect structure of the catalyst was further characterized using EPR spectroscopy, and the results are shown in Figure 2i. The presence of the unpaired electrons on the catalyst surface and the substrate is characterized using EPR, the intensity of which depends on the number of unpaired electrons and the electron mobility.⁴⁷ The type of relative defect and concentration can be obtained from different g-values and signal intensities. The EPR Lorentzian line centers of $\text{Ni}_{20}\text{Zn}_1\text{@NC}$, $\text{Ni}_{20}\text{Zn}_1\text{@C}$, and Ni@NC catalysts are obtained at g-values of 2.0033, 2.0030, and 2.0035, respectively. $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ exhibits the higher peak value than that of the other two, indicating the presence of maximum unpaired electrons. The results established that the

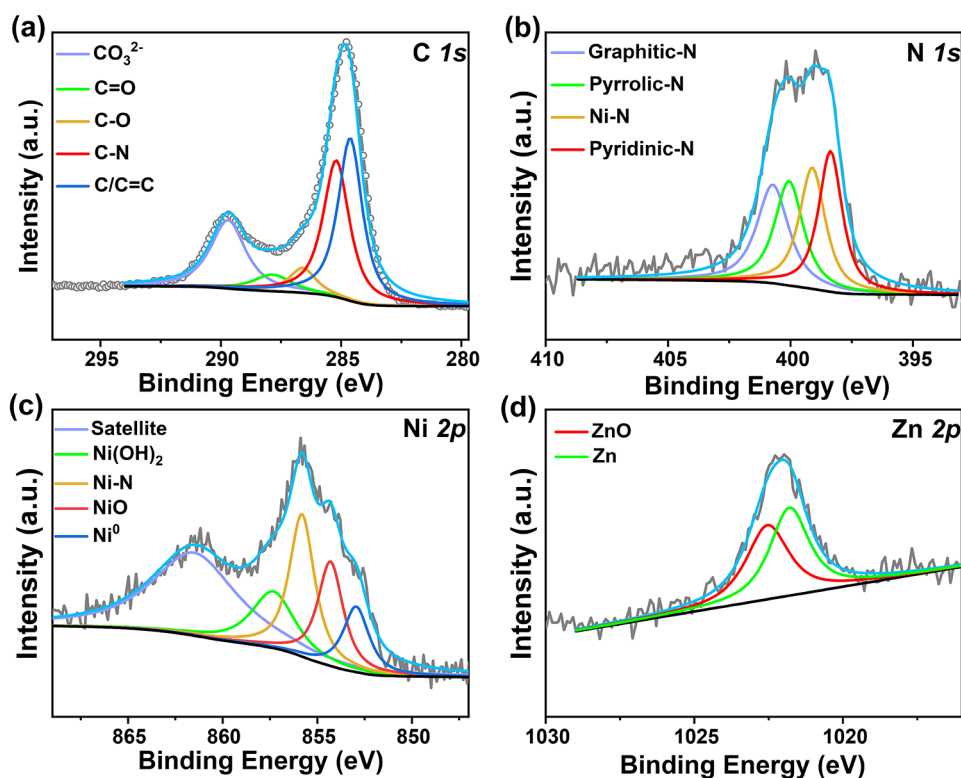


Figure 3. XPS spectra of the $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ catalyst: (a) C 1s spectrum, (b) N 1s spectrum, (c) Ni 2p spectrum, and (d) Zn 2p spectrum.

introduction of Zn alters the electronic environment of Ni and carbon carrier, and increase the number of oxygen and nitrogen vacancies in the catalyst.²¹ In addition, N doping also increases nitrogen vacancies and improves the defect structure of the catalyst, which is in agreement with the Raman and HRTEM results. The defective structure of the composite catalyst was generated due to the synergistic effect of Zn doping and N-doped carbon carrier as indicated by Raman spectroscopy and EPR.

The surface chemical composition and chemical state of the $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ catalyst were further analyzed using XPS in the range of 0–1350 eV. Ni 2p, Zn 2p, N 1s, and C 1s signals are observed in the survey scan spectra (Figure 3), which confirms the presence of Ni, Zn, N, and C on the surface of the $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ catalyst. The peak at 284.6 eV in C 1s is ascribed to the C/C=C bond, while the peaks at 285.2 and 287.9 eV correspond to C–N and C=O bonds. Additionally, the C 1s profile exhibits a peak for CO_3^{2-} with a binding energy of 289.7 eV that is associated with the fibrous material produced by Na_2CO_3 .⁴⁸ The N 1s spectra at binding energies of 398.4, 400.1, 400.7, and 399.1 eV are attributed to pyridinic nitrogen, pyrrolic nitrogen, graphitic nitrogen, and Ni–N bonds, respectively (Figure 3b).⁴⁹ In particular, the peak for the Ni–N bond is observed in the Ni 2p spectrum at 855.8 eV. This is because the Ni ions coordinate with the amine groups of the lignin derivatives during the synthesis of the catalyst, resulting in a homogeneous dispersion of the active Ni in the lignin-derived carbon. The Ni 2p peak exhibited a distinct spin–orbit component, and the Ni $2p_{3/2}$ area had both the primary peak and a satellite peak located close to 861.5 eV in Figure 3c. Ni $2p_{3/2}$ peaks at 857.3, 854.3, and 852.9 eV are attributed to Ni(OH)_2 , NiO, and Ni^0 , respectively.^{5,50} The Zn 2p spectrum (Figure 3d) indicated that Zn is dispersed in the

carbon support in its metallic state (1021.8 eV) and ZnO (1022.5 eV).

XAFS spectra were conducted to investigate the atomic configuration of the $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ catalyst. The normalized intensity of Ni K-edge and its Fourier-transform k^3 -weighted spectra in the R space for the Ni foil, Ni@NC, and $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ is shown in Figure S5. In comparison with metal Ni, $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ and Ni@NC catalysts exhibit a stronger intensity of the white peak and the blue shift of the edge absorption, indicating the increase of oxidation state generated by the presence of the coordination with N/O. The results demonstrate a reduced Ni–Ni coordination with the doping of N and Zn, and the intensity of $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ at the radial distance of 2.50 Å ascribed to the Ni–Ni bond was lower than that of Ni@NC, suggesting as a result of the formation of Ni–Zn.²³

Aqueous Ethanol Upgrading to Higher Alcohols. The effect of $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ catalysts prepared with varied degrees of amination-modified lignin on the conversion of ethanol to higher alcohols is shown in Figure 4. The conversion of ethanol and yield of higher alcohols are 70.8 and 29.2%, respectively, in the ethanol coupling reaction using the $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ catalyst prepared from carboxymethylated lignin when applied in the ethanol coupling reaction. The catalytic activities of $\text{Ni}_{20}\text{Zn}_1\text{@NC}$ catalysts prepared from 1/4 aminated modified lignin are similar (ethanol conversion of 71.3% and yield of higher alcohols of 28.4%). The yields of each higher alcohol in the liquid product are comparable with those of the control. When the degree of lignin amination is increased from 1/4 to 1/2, the ethanol conversion has the highest value of 75.2%, and the yield of higher alcohols increases significantly from 28.4 to 41.7%. Both ethanol conversion and the yield of higher alcohols show a decreasing

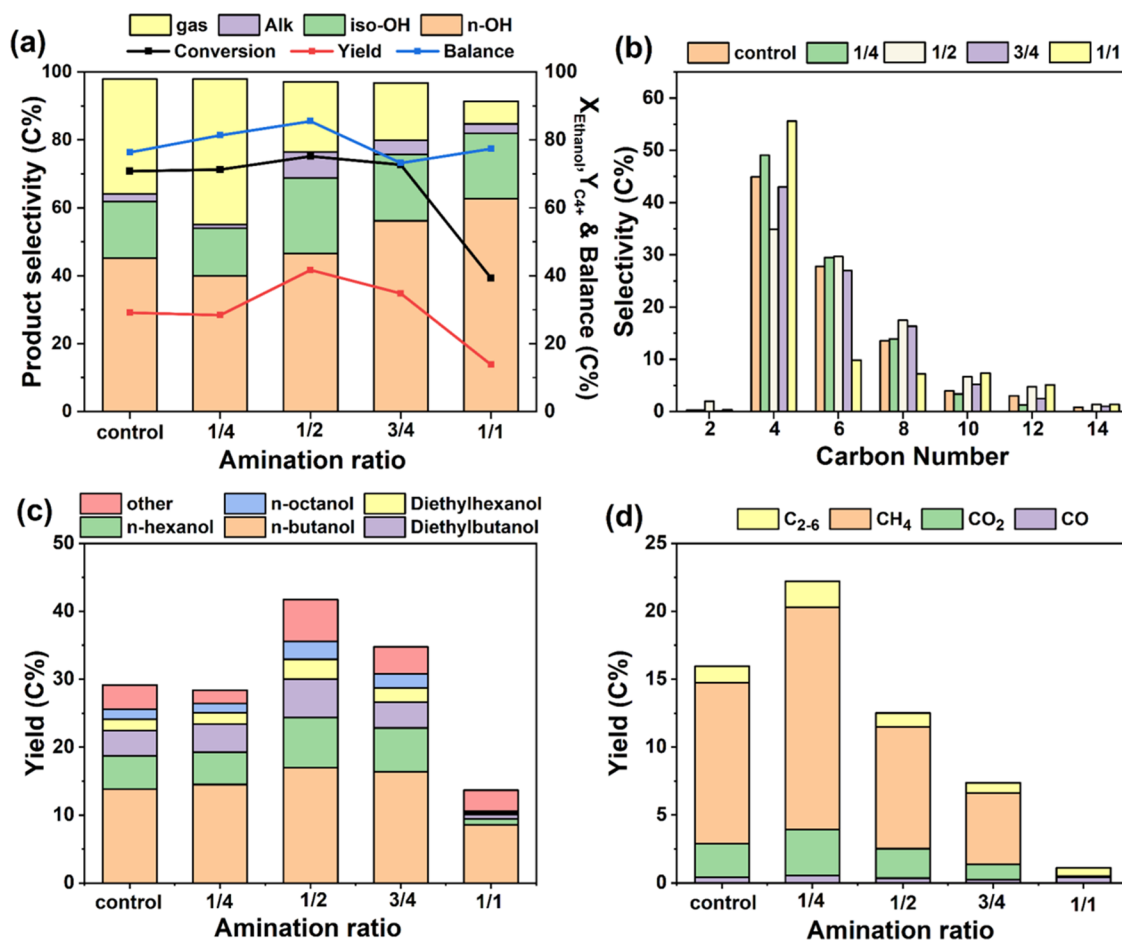


Figure 4. Effect of Ni₂₀Zn₁@NC catalysts prepared from modified lignin with various degrees of amination on the conversion of ethanol to higher alcohols: (a) ethanol conversion, higher alcohols yield, product selectivity, and carbon balance, (b) carbon distribution of higher alcohol products, (c) yield of major liquid products, and (d) distribution of gaseous products.

trend as the degree of lignin amination increases to 1/1. Ethanol conversion decreases from 75.2 to 39.3%, while the yield of higher alcohols decreases from 41.7 to 13.8%. There is no significant increase in ethanol conversion when the degree of lignin amination is increased from 0 to 3/4, owing to the minor difference in the particle size of the metal clusters in the prepared Ni₂₀Zn₁@NC catalysts. Although the metal particle size of the catalyst synthesized with the 1/1 amination-modified lignin is 7 nm (Figure S4) with an increased metal dispersion, the ethanol conversion and yield of higher alcohols (13.8%) are significantly below that of the catalyst prepared using carboxymethylated lignin. This is attributed to the presence of a lower amount of doped Zn and the lowest degree of defect in the catalyst prepared using the 1/1 amination-modified lignin. The metal crystal defects promote the hydrogenation and dehydrogenation of CH₃CH₂OH molecules and other intermediates, while lignin-derived carbon can disperse the metal, and the defect structures of carbon materials can facilitate the adsorption of H₂ molecules, CH₃CH₂OH molecules, and other reaction intermediates during the ethanol coupling reaction. The carbon layer efficiently enhances the electron-conducting ability in internal materials and restrains active particle aggregation. However, the thick carbon layer reduces the contact between the reactants and mass transfer of intermediates.^{26,51} Compared with commonly used homogeneous and heterogeneous catalysts for ethanol coupling, the as-synthesized Ni₂₀Zn₁@

NC catalysts exhibited excellent ethanol conversion and comparable higher alcohols yield, as shown in Table S4.

The carbon distributions of the catalytic products of Ni₂₀Zn₁@NC catalysts prepared with various degrees of amination-modified lignin are shown in Figure 4b. The carbon number distribution of the higher alcohols spans from C₄ to C₁₄ and the amount of each product decreases from C₄ to C₁₄. The catalyst prepared from 1/2 amination-modified lignin exhibits the slowest decreasing trend in the product, with promising conversions to the C₈–C₁₄ higher alcohol products. The predominant product of the catalyst prepared using the 1/1 amination-modified lignin is *n*-butanol; it cannot convert *n*-butanol to the corresponding aldehyde, resulting in low content of C₆₊ alcohols. Figure 4c,d shows the liquid product distribution and gaseous product distribution, respectively, of Ni₂₀Zn₁@NC catalysts prepared with various degrees of amination-modified lignin applied to the ethanol coupling reaction. The main liquid products are *n*-butanol, *n*-hexanol, diethylbutanol, diethylhexanol, and *n*-octanol, while the main gaseous products are CH₄ and CO₂. The selectivity of the gaseous products reduces with increasing amination of lignin, indicating that increasing amination inhibits the C–C bond cleavage.

NiZn@NC catalysts with varied Ni/Zn ratios were applied in the ethanol coupling reaction to study the effect of Zn doping on the product distribution, shown in Figure 5. Ethanol conversions of Ni@NC and Zn@NC catalysts are 81.1 and

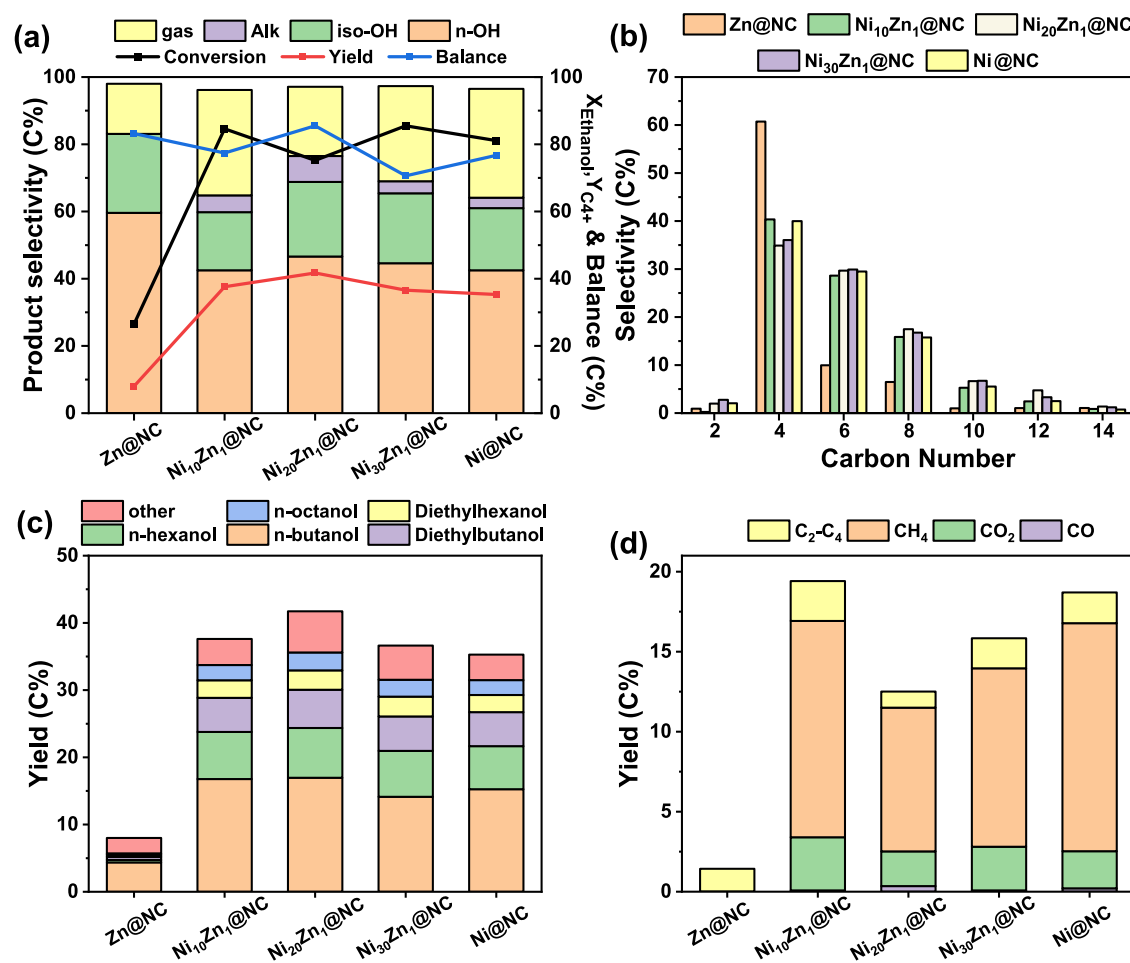


Figure 5. Effect of varied Ni/Zn ratios in the catalyst: (a) Ethanol conversion, higher alcohols yield, product selectivity, and carbon balance, (b) carbon distribution of higher alcohol products, (c) yield of major liquid products, and (d) distribution of gaseous products.

26.5%, respectively (Figure 5a). This proves that Ni actively participates in dehydrogenation and bond cleavage of the reaction. Moreover, Zn is responsible for the intermediate coupling reaction. Despite the high ethanol conversion of 81.1% over the Ni@NC catalyst, the formation of gases, mainly CH₄, CO₂, CO, and C₂–C₆ alkane with yields of 14.3, 2.3, 0.2, and 1.9%, respectively, was observed (Figure 5d). Zn doping inhibits C–C and C–O bond cleavage and reduces the formation of byproduct gases such as CH₄. On increasing the amount of Zn in Ni/Zn ratio to 20:1, the ethanol conversion decreases and the gas formation decreases from 32.4 to 20.6%. However, the yield of higher alcohols increased from 35.3 to 41.7%. A further increase of Zn in the Ni/Zn ratio from 20:1 to 10:1 is accompanied by increased gas and a corresponding decrease in the yield of higher alcohols to 37.6%. The yields of higher alcohols in the aqueous ethanol coupling reaction catalyzed by various catalysts are in the following order: Ni₂₀Zn₁@NC > Ni₁₀Zn₁@NC > Ni₃₀Zn₁@NC > Ni@NC, which is consistent with the Zn-doping effect on the defect degree of the as-synthesized carbon material determined by Raman spectroscopy.

The effect of the Ni/Zn ratios in the NiZn@NC catalysts on the carbon distribution of higher alcohols is shown in Figure 5b. For various Ni/Zn molar ratios in the NiZn@NC catalysts, the distribution of higher alcohols spans from C₄ to C₁₄, and product selectivity tends to decrease with the increasing carbon number of the higher alcohols. The Ni₂₀Zn₁@NC catalyst has

the lowest yield of *n*-butanol (C₄) when applied to ethanol coupling and higher yields of higher alcohols in the range of C₆–C₁₂, enhancing the quality of the product as aviation fuel. The main liquid products of the aqueous ethanol coupling reaction are *n*-butanol, *n*-hexanol, diethylbutanol, diethylhexanol, and *n*-octanol (Figure 5c) in decreasing yields, indicating that higher alcohols are more likely to be formed as branched alcohols as the carbon chain increases. Overall, the yields of higher alcohols over Ni₁₀Zn₁@NC, Ni₂₀Zn₁@NC, Ni₃₀Zn₁@NC, and Ni@NC catalysts are all above 35%, while a wide range of higher alcohols yields are obtained for Ni₂₀Zn₁@NC catalysts prepared from modified lignin with various degrees of amination, indicating that the N content has a major effect on the performance of the catalysts.

In addition, the effects of the coordination pH (Figure S6) during the preparation of Ni₂₀Zn₁@NC catalysts and the effects of various reaction initiation pressures (Figure S7) on the conversion of ethanol to higher alcohols over Ni₂₀Zn₁@NC catalysts were investigated. pH affects the coordination of the carboxymethylated and aminated lignin with the NiZn clusters and consequently the catalytic activity of the catalyst (Figure S6). The highest ethanol conversion (75.25%) and highest yield of higher alcohols (41.7%) are achieved when the metal-complexation pH value is 6.7.

As the initial pressure of H₂ increased from atmospheric pressure (0.1 MPa) to 2 MPa (Figure S7), the conversion of ethanol gradually increases from 75.2 to 81.7%; however, the

yield of the higher alcohols gradually decreases from 41.7 to 34.8%. This is attributed to the gradual increase in gas selectivity in the byproduct of the ethanol Guerbet coupling reaction as the H_2 initial pressure increases. The increased pressure of H_2 promotes the hydrogenation process and facilitates the formation of CH_4 and CO_2 from ethanol and intermediates over the $Ni_{20}Zn_1@NC$ catalyst,⁵² leading to an increase in gas selectivity. Additionally, the basic site (NaOH) is further consumed by CO_2 , decreasing the yield of C_{8+} alcohols.

Recycle Performance. The $Ni_{20}Zn_1@NC$ catalyst was used over four cycles of aqueous ethanol coupling to evaluate its stability, as shown in Table 1. The ethanol conversion

Table 1. Reusability of the Spent $Ni_{20}Zn_1@NC$ Catalyst in Aqueous Ethanol Upgrading

use cycle	mass (g)	conversion (C%)	yield _{C₄+} (C%)	product selectivity (C%)		
				<i>n</i> -OH	iso-OH	balance (C%)
1	0.50	75.2	41.7	46.6	17.2	77.1
2	0.24	57.8	27.5	51.7	23.4	71.5
3	0.17	66.6	33.7	55.8	23.3	76.1
4	0.14	65.2	30.3	54.1	21.3	75.1

decreases slightly from 75.2 to 65.2%, and the yield of higher alcohols also decreases from 41.7 to 30.3% after four reaction cycles. However, the selectivity to higher alcohols increases, while the formation of byproduct gases decreases using the recycled catalyst. The structure and morphology of the spent catalyst after the fourth cycle were characterized using SEM,

TEM, XRD, and XPS to study its stability (Figure 6). The nanometallic particles remained uniformly dispersed on carbon after four reaction cycles (Figure 6e,f) and had the same morphology as the freshly prepared catalyst. The Zn content in the spent $Ni_{20}Zn_1@NC$ catalyst was leached and decreased from 0.82 to 0.28%, while the primary active Ni remained stable, according to the ICP-OES as shown in Table S3. With the leaching of Zn, more Ni sites were exposed on the surface, which explained the increment of activity based on per gram of catalyst. XRD patterns and XPS spectra of fresh and spent $Ni_{20}Zn_1@NC$ catalysts are shown in Figure 6. Ni–N–C and Ni^{2+} state are the main state of $Ni_{20}Zn_1@NC$ catalyst, Ni–N–C usually promoted the hydrogenation and dehydrogenation, and Ni^0 is oxidized to Ni^{2+} during the ethanol coupling reaction, which is beneficial for dehydrogenation and following hydrogenation reactions.²² As the amount of Zn reduced in the recycled catalysts, a limited decrease in the yield of higher alcohols was noted. The strategy to improve the stability of Zn doping in NiZn bimetallic catalyst will be further investigated.

Catalytic Mechanism and Discussion. The reaction mechanism for the conversion of ethanol to CH_4 and *n*-butanol over Ni catalysts and the effect of Zn on the reaction selectivity were investigated using DFT calculations. The possible pathways and reaction energy profiles for the conversion of ethanol to the byproduct CH_4 over the Ni(111) surface are shown in Figure 7a. The configurations of intermediates and transition states during ethanol decomposition to CH_4 and ethanol coupling to *n*-butanol on Ni(111) is displayed in Figures S8 and S9. The conversion of ethanol to CH_4 involves C–H, C–O, and C–C bond breaking steps, among which the C–C bond breaking of CH_3CO^*

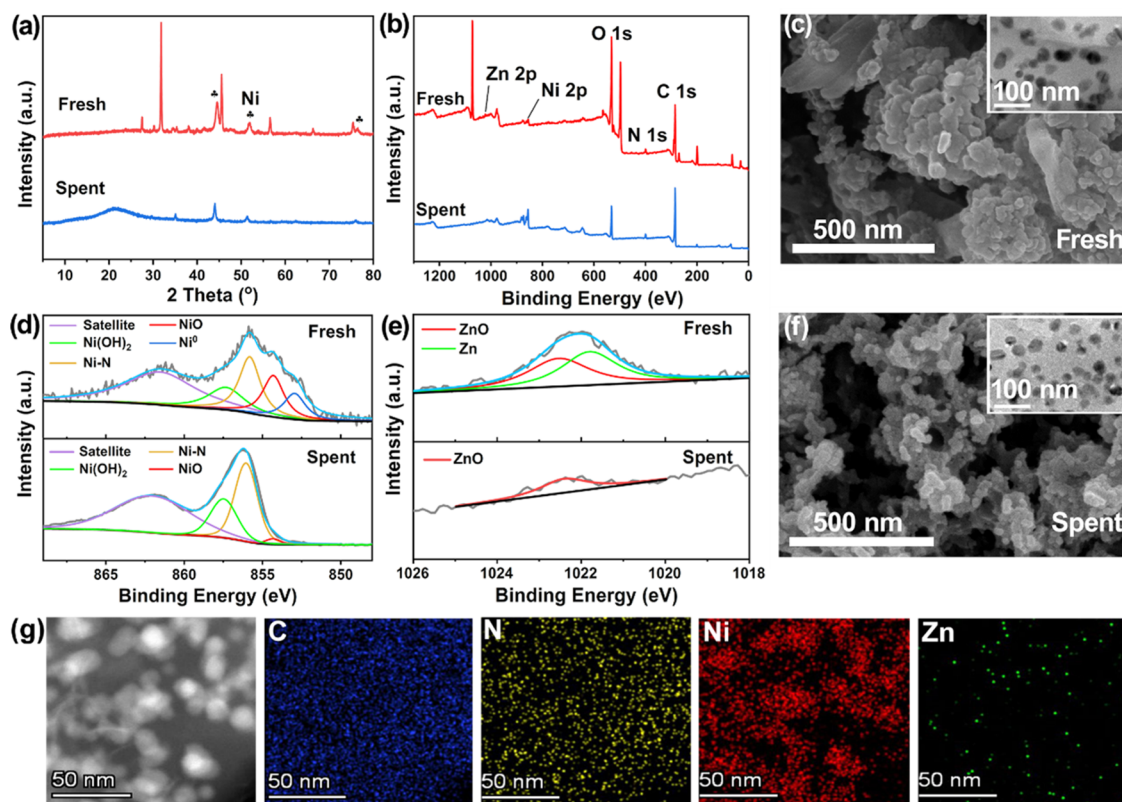


Figure 6. (a) XRD, (b, d, e) XPS, (c, f) TEM results of the fresh and spent $Ni_{20}Zn_1@NC$ catalyst, and (g) HAADF-STEM image, and EDS elemental map of C, N, Ni, and Zn for the spent $Ni_{20}Zn_1@NC$ catalyst.

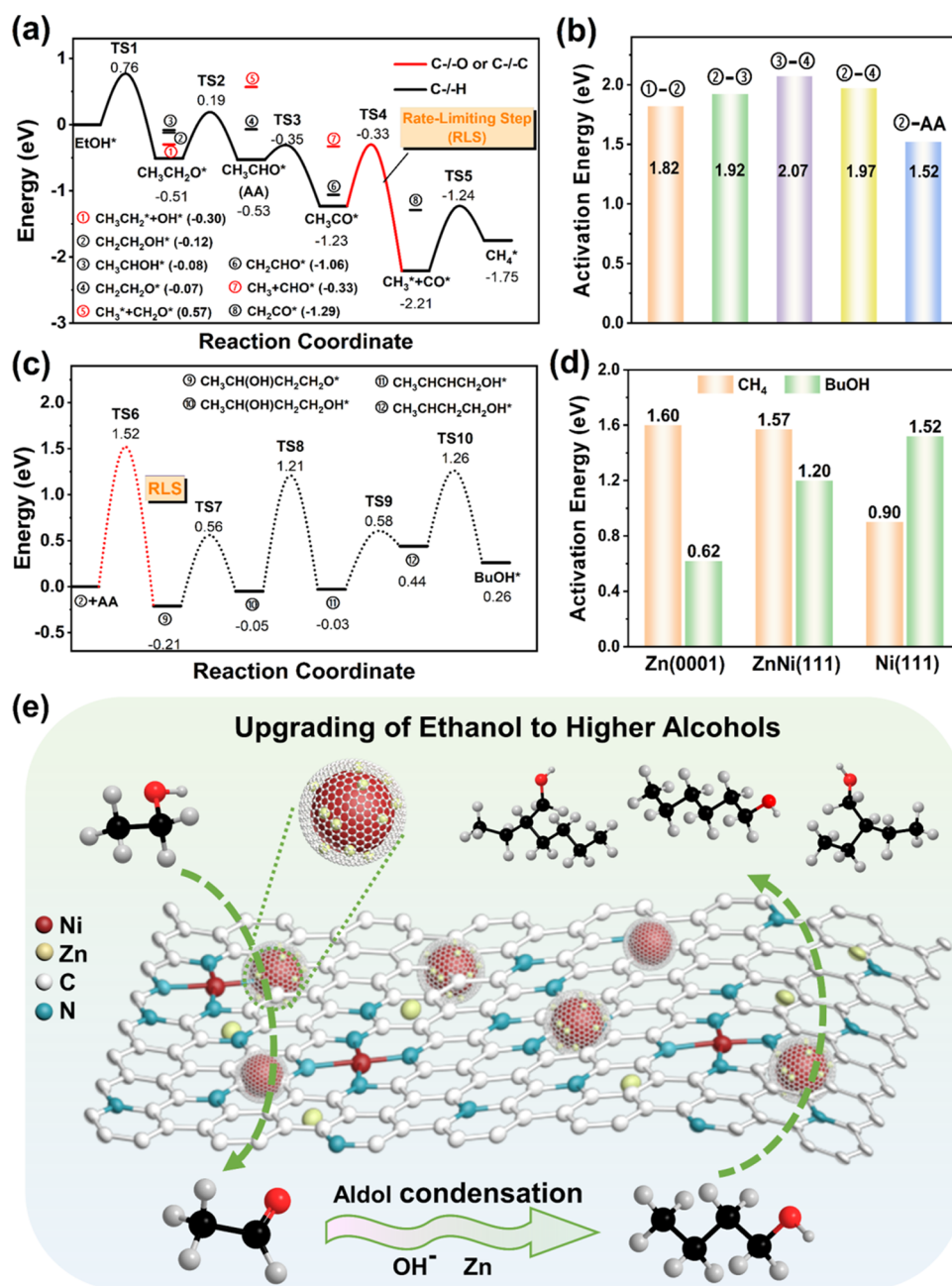


Figure 7. (a) Reaction energy curves for the conversion of ethanol to CH₄ over the Ni(111) catalyst, (b) activation energies for the C–C coupling reaction, (c) energy curves for the conversion of ethanol to *n*-butanol over the Ni(111) crystal plane, (d) activation energy for the rate-limiting step of CH₄ formation and the rate-limiting step of *n*-butanol formation over the Zn(0001), Ni(111), and ZnNi(111) crystal planes, and (e) reaction pathway of ethanol to higher alcohols.

intermediate to form CH₃* and CO* intermediates is the rate-limiting step (RLS) in the formation of CH₄. This RLS (TS4 in Figure 7a) requires an activation energy of 0.90 eV. The intermediates produced during ethanol decomposition can undergo C–C coupling to produce higher alcohols. Various possible pathways of C–C bond coupling were explored, with the coupling reaction of CH₂CH₂OH* (②) and CH₃CHO* (AA) intermediates requiring the lowest activation energy (1.52 eV) and being the most energetically favorable C–C bond coupling pathway (Figure 7b). Based on this, the reaction mechanism of *n*-butanol production over Ni(111), with the addition of CH₂CH₂OH* and CH₃CHO* as the starting step, was further investigated, and the reaction energy

profile is shown in Figure 7c. In particular, the C–C coupling reaction of CH₂CH₂OH* and CH₃CHO* requires the highest activation energy (TS6, 1.52 eV) along the reaction course of *n*-butanol formation and is determined as the rate-limiting step. The activation energies of the rate-limiting steps of CH₄ and *n*-butanol formation over Zn(0001) and Ni(111) were compared to investigate the role of Zn in ethanol conversion, and the results are shown in Figure 7d. All intermediate and transition-state configurations are shown in Figures S8–S10. The activation energy required for the C–C bond cleavage in the RLS during CH₄ formation is 1.60 eV over Zn(0001), which is 0.70 eV higher than that observed over Ni(111), whereas the C–C bond coupling RLS during *n*-butanol formation on

Zn(0001) requires an activation energy of 0.9 eV lower than that on Ni(111) (0.62 eV vs 1.52 eV). We further calculated the barriers of the two RLSs on the ZnNi(111) surface. It turned out that Zn doping significantly elevates the C–C bond cleavage (TS4) barrier by 0.67 eV, while reducing the C–C bond coupling barrier by 0.32 eV. Therefore, the Zn-doped catalyst effectively inhibits the C–C bond cleavage and production of gases such as CH₄ during the ethanol coupling reaction, as well as promotes the conversion of ethanol via C–C bond coupling to higher alcohols such as *n*-butanol, which is consistent with the experimental results of the ethanol coupling reaction.

The reaction pathway of aqueous ethanol to higher alcohols over the Ni₂₀Zn₁@NC catalyst is illustrated in Figure 7e. The formation of higher alcohols follows a typical Guerbet reaction mechanism, as illustrated in Figure S11. Ethanol produces intermediates such as CH₃CH₂O*, CH₃CHO*, CH₃CO*, and CH₂CH₂OH* in the presence of the catalyst (Figure S11). Among the intermediates, CH₂CH₂OH* and CH₃CHO*, which are the easiest to undergo C–C bond coupling, can react via Aldol condensation to form CH₃CH(OH)-CH₂CH₂O*, which is further dehydrated and hydrogenated to form the target product *n*-butanol. The C–C bond cleavage occurs in various intermediates, producing byproducts such as CH₄, CO, and CO₂, leading to a lower selectivity of higher alcohols. The Zn doping efficiently inhibits C–C bond cleavage and the production of gaseous byproducts; thus, the Zn-doped catalysts boost the conversion of C–C bond coupling of ethanol to higher alcohols.

CONCLUSIONS

NiZn@NC catalysts with abundant defective structures were prepared using alkali lignin from pulp and paper waste streams as a carbon source, followed by doping with Zn. The alkali lignin was modified via carboxymethylation and amination to increase its metal coordination and nitrogen content. An ethanol conversion of 75.2% and a higher alcohols yield of 41.7% were achieved with the Ni₂₀Zn₁@NC catalyst at an amination reagent/lignin mass ratio of 1:2, and the catalyst showed excellent stability. Doping with Zn/N increased the number of electrons on Ni and facilitated the generation of defective structures. This effectively inhibited C–C and C–O bond cleavage and improved the selectivity of the target products during the ethanol coupling reaction. The defective structure of the catalyst and the degree of dispersion of the metal clusters were improved by varying the doping content of N in the alkali lignin modification process, thereby increasing the conversion and yield of the aqueous ethanol coupling reaction. The Ni₂₀Zn₁@NC catalyst showed promising applications in the efficient conversion and preparation of higher alcohols.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.2c02440>.

Results of ethanol coupling reactions on various catalysts, catalyst models used in DFT calculations, catalyst characterization, supplementary data of ethanol coupling, reaction pathway of ethanol to higher alcohols, and configurations of intermediates and transition states in DFT calculations (PDF)

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Notes

The authors declare no competing financial interest.

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